Week 10

House Sales Analytics Dashboard in Power BI

King County House Sales Dataset (Preprocessed from Week 9)

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1. Introduction

Real-estate managers need to understand how property prices change with location, size and quality, and how sales are spread across the county. A table of thousands of rows is hard to read, so an interactive dashboard is a better way to explore the data. In this project I built the House Sales Analytics Dashboard in Power BI Desktop using the King County House Sales dataset that I cleaned and prepared in Week 9.

The dashboard combines summary numbers, a location map, price charts and slider controls on one page. A manager can see the overall market in a few seconds and then look closer at areas, grades, conditions and waterfront houses.

1.1 Problem Statement

Real-estate managers want to understand property prices, locations, house characteristics and sales patterns using an interactive Power BI dashboard.

1.2 Objectives

- Formulate real estate DAX measures: Total Properties Sold, Avg House Price, Median Price, Avg Living Area and Avg Price/SqFt.
- Construct a spatial visualization using a map based on latitude and longitude.
- Evaluate price impacts driven by property grade, condition and waterfront availability.
- Set up numeric range parameters for dynamic price and square footage filtering.

1.3 Tasks and Deliverables

- Build an interactive geographic map visual color-coded by sales price.
- Create price distribution charts segmented by bedrooms, bathrooms and waterfront status.
- Implement dynamic parameter sliders for square footage filtering.
- Deliver the House Sales Analytics Dashboard.

2. Data and Tools

The data comes from the King County House Sales dataset. In Week 9 the file was imported into Power Query, cleaned, and loaded into the Power BI data model as the table Housing_Clean. Rows with zero bedrooms or zero bathrooms and the unrealistic 33-bedroom record were removed, and the sale date was converted to a proper Date column.

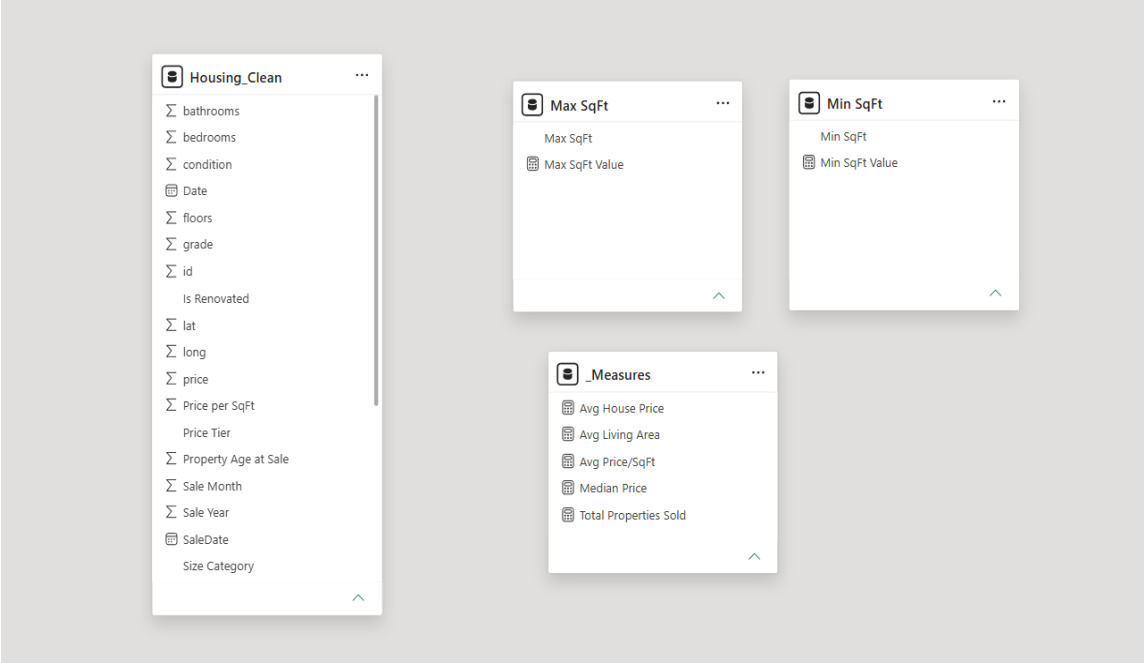
I created the table contains the derived columns : Sale Year, Sale Month, Property Age at Sale, Price per SqFt, Total Area, Is Renovated, Size Category and Price Tier. These columns are used directly in this week's visuals, for example Price Tier is used to color the map and Size Category is used in the treemap.

2.1 Tools Used

- Power BI Desktop for the data model, DAX measures, parameters and visuals.
- DAX (Data Analysis Expressions) for measures and calculated columns.
- Map visual (Bing map) for the geographic view using latitude and longitude.

2.2 Model Organization

To keep the data model tidy, all measures were stored in a separate table called _Measures, while the raw and derived columns stay in Housing_Clean. This separation makes the measures easy to find and avoids mixing calculations with data columns.



This figure shows the model, which has four tables. Housing_Clean holds the cleaned house data and the derived columns from Week 9, such as Price Tier, Size Category, Sale Year, Sale Month, Property Age at Sale and Price per SqFt. The table _Measures holds only the five DAX measures: Avg House Price, Avg Living Area, Avg Price/SqFt, Median Price and Total Properties Sold.

The tables Max SqFt and Min SqFt were created automatically by the numeric range parameters. Each has one value column and one measure, Max SqFt Value and Min SqFt Value, which return the number selected on the slider. These parameter tables have no relationship lines to Housing_Clean on purpose, since they only supply a selected value.

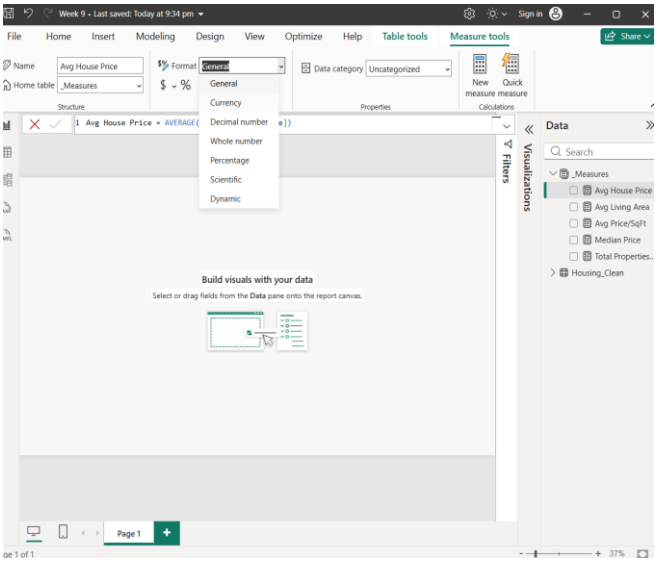
2.3 DAX Measures Created

Measure	DAX formula	Purpose
Total Properties Sold	COUNTROWS(Housing_Clean)	Counts the number of houses sold
Avg House Price	AVERAGE(Housing_Clean[price])	Shows the average sale price
Median Price	MEDIAN(Housing_Clean[price])	Middle price, not affected by very expensive houses
Avg Living Area	AVERAGE(Housing_Clean[sqft_living])	Average living space in square feet
Avg Price/SqFt	AVERAGE(Housing_Clean[Price per SqFt])	Average price paid per square foot

3. Implementation and Screenshots

This section explains each step of the build with the screenshot taken at that stage. The measures were created and formatted first, then the extra column, the parameters and the report page were built.

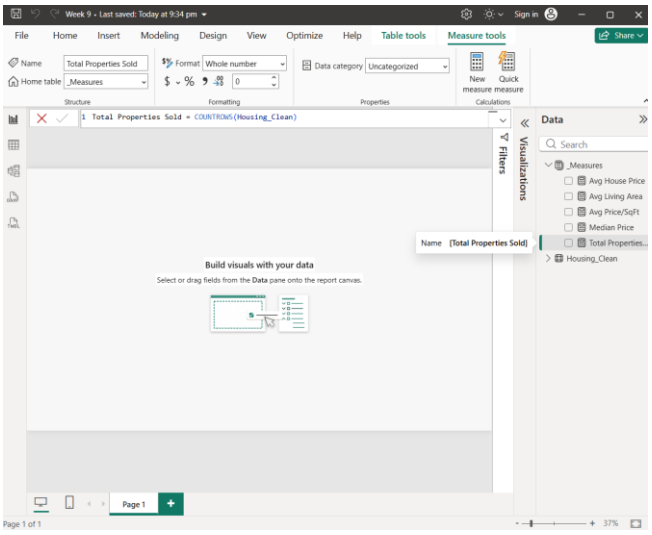
3.1 Formatting the Avg House Price Measure



After a measure is created, its display format must be set so the numbers are readable. This screenshot shows the Measure tools tab in Power BI with the measure Avg House Price selected in the _Measures table. The Format drop-down is open and lists General, Currency, Decimal number, Whole number, Percentage, Scientific and Dynamic.

For price measures I selected Currency, so values appear as dollar amounts such as $540K instead of a raw number with many decimals. The formula bar at the top shows the DAX expression for this measure, AVERAGE(Housing_Clean[price]).

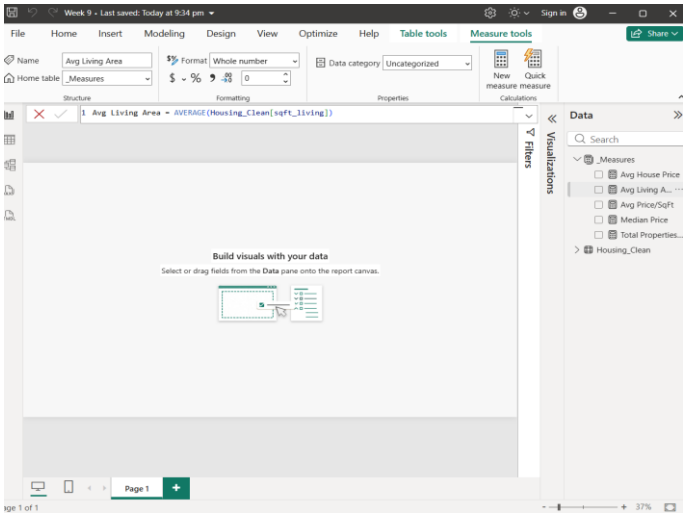
3.2 Total Properties Sold



Total Properties Sold counts how many rows are in the Housing_Clean table, which means how many house sales are included in the analysis. COUNTROWS is the right function here because every row is one sale.

The measure is stored in _Measures and its format is set to Whole number, so the card shows a clean count with no decimals. This measure is the first KPI on the dashboard and gives the size of the data being analyzed.

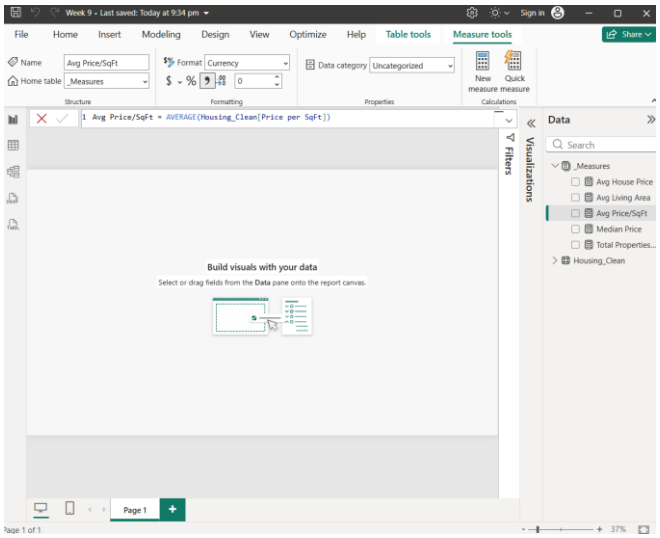
3.3 Avg Living Area



Avg Living Area calculates the average value of the sqft_living column, which is the interior living space of a house in square feet. It helps managers understand the typical size of the houses in the market.

Because square footage is a whole-number idea, the format is Whole number with no decimals. On the dashboard card the value is shown in thousands, about 2K square feet.

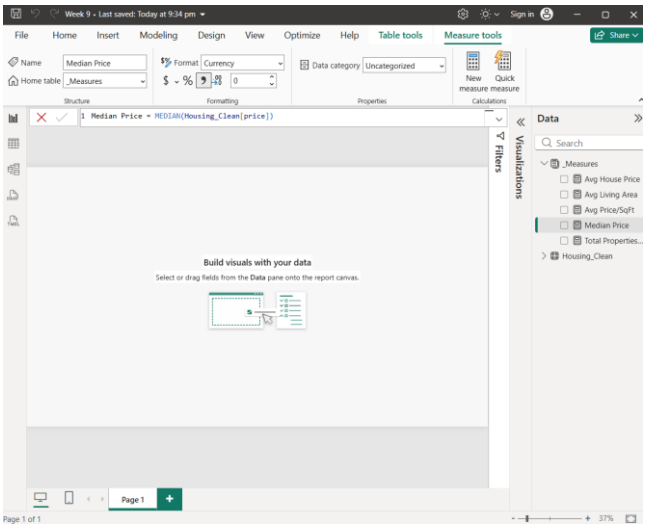
3.4 Avg Price per SqFt



Avg Price/SqFt uses the Price per SqFt column that was created in Week 9 as price divided by living area. Averaging it gives the typical price paid for each square foot of living space.

This measure allows fair comparison between small and large houses, because a large house is naturally more expensive. The measure is formatted as Currency and appears on the dashboard as about $264 per square foot.

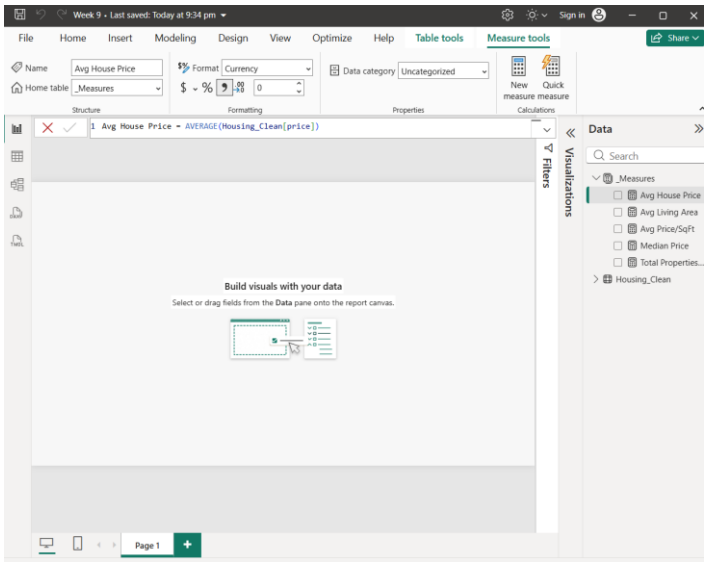
3.5 Median Price



Median Price returns the middle sale price when all prices are sorted. Unlike the average, the median is not pulled upward by a few very expensive houses, so it shows what a typical buyer pays.

On the dashboard the median is about $450K, which is lower than the average of about $540K. This gap shows that a small number of high-priced properties raise the average.

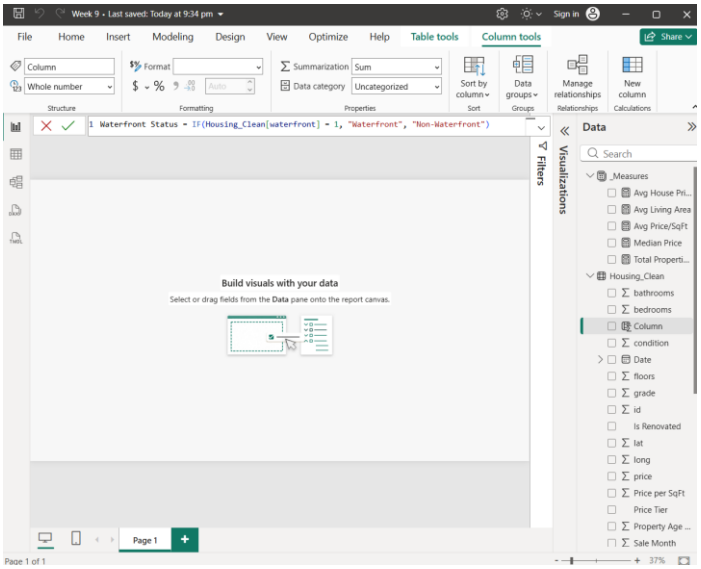
3.6 Avg House Price



Avg House Price takes the mean of the price column. It is the main price indicator of the dashboard and is also used as the value in most charts, such as price by grade, bedrooms, bathrooms and condition.

The Home table is set to _Measures and the format is Currency with no decimals. Using one measure in many charts keeps all visuals consistent.

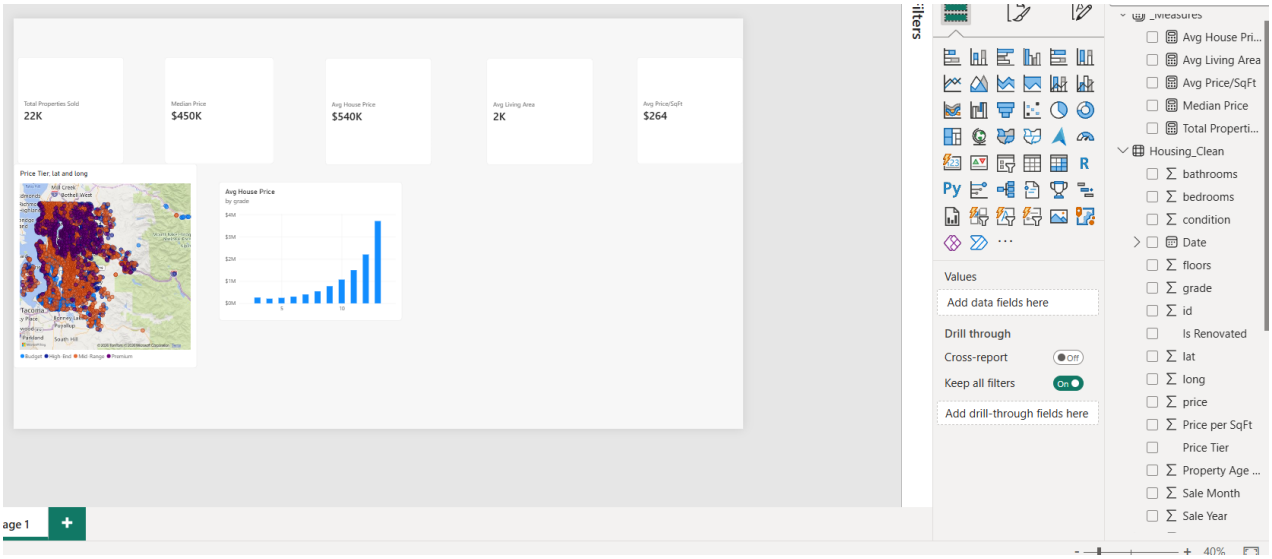
3.7 Waterfront Status Calculated Column



In the data, the waterfront field is stored as 0 or 1, which is not easy to read in a chart. I created a calculated column named Waterfront Status in the Housing_Clean table with the formula IF(Housing_Clean[waterfront] = 1, "Waterfront", "Non-Waterfront").

The screenshot shows the Column tools tab with the formula in the formula bar and the new column listed in the Data pane. This column makes the waterfront comparison chart easier to understand because it shows clear labels instead of 1 and 0.

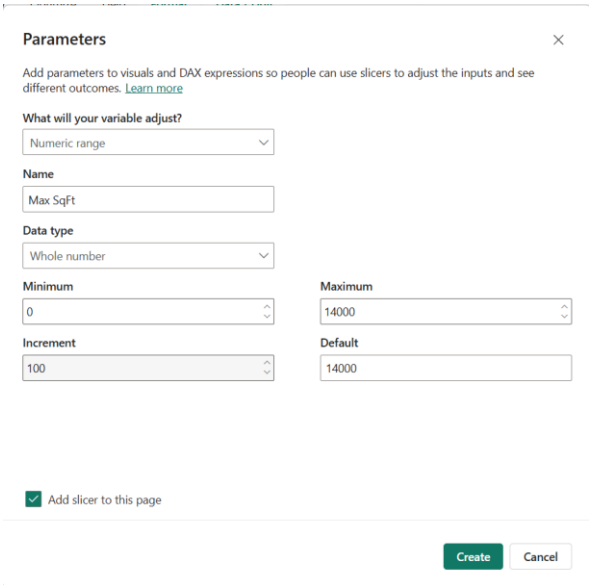
3.8 Early Draft with Cards, Map and Grade Chart



This screenshot shows an early stage of the build. The five measure cards are placed in a row, the map is added below them with the Price Tier legend, and the first chart, average house price by grade, is placed next to the map.

The Visualizations pane on the right lists the available visual types and the Data pane lists the fields from Housing_Clean and _Measures. The chart shows that average price increases with grade, and the rise is steepest at the highest grades.

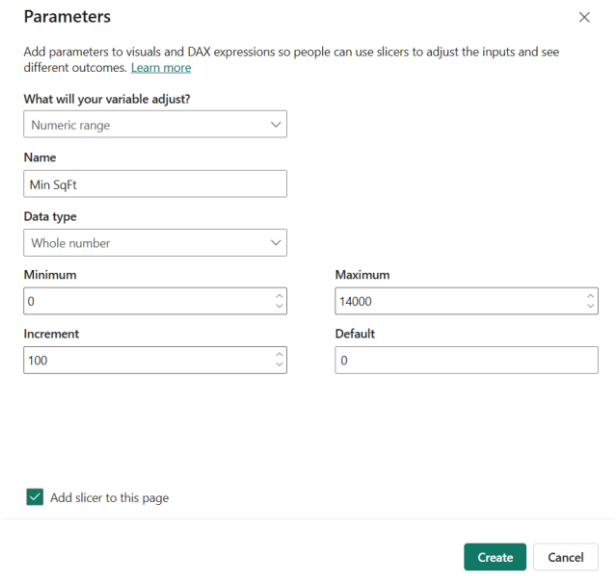
3.9 Max SqFt Parameter



A parameter lets a viewer change a value with a slider. In the Modeling tab I used New parameter and chose Numeric range. The Max SqFt parameter has the data type Whole number, a minimum of 0, a maximum of 14000, an increment of 100 and a default of 14000.

The option Add slicer to this page was ticked, so Power BI created the slider automatically. Power BI also creates a hidden table and a measure that returns the selected value, which can be used inside DAX formulas to filter the data by living area.

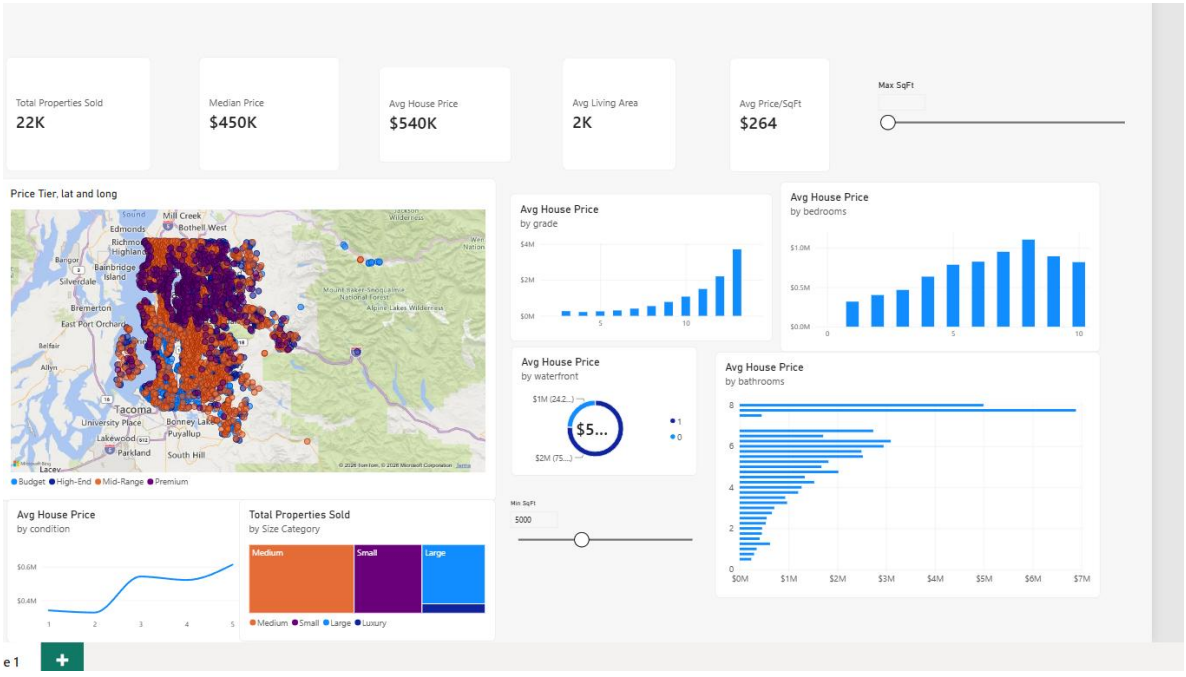
3.10 Min SqFt Parameter



The Min SqFt parameter is created in the same way as Max SqFt. Its range is 0 to 14000, the increment is 100 and the default is 0, so by default no house is excluded.

Together, the Min SqFt and Max SqFt sliders let a manager set a lower and an upper limit for living area, for example to look only at houses between 2,000 and 3,000 square feet. This fulfils the dynamic parameter slider requirement of the task.

3.8 First Dashboard Draft

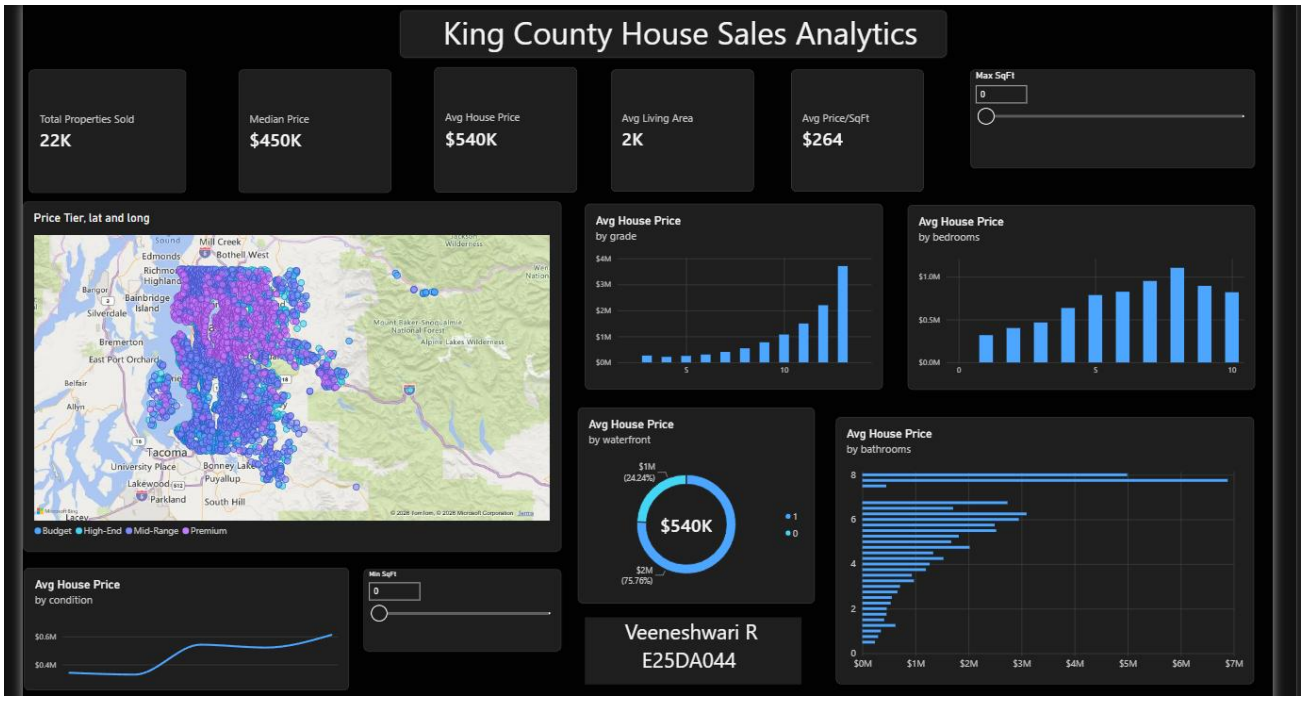


This is the first full draft of the report page. The top row has the five KPI cards, and a Max SqFt slider is placed at the right. Below them is the map, followed by the column charts for average price by grade and by bedrooms, a donut chart for waterfront, a bar chart for bathrooms, a line chart for condition and a treemap of properties by size category.

The map uses latitude and longitude and colors the points by Price Tier, so Budget, Mid-Range, High-End and Premium houses can be seen at a glance. A Min SqFt slider is shown

near the bottom. This draft shows the overall structure of the dashboard before the dark styling was applied.

3.12 Final House Sales Analytics Dashboard



The final dashboard uses a dark theme with a title banner, KPI cards, the map, the slider controls and the price charts. The cards show about 22K properties, a median price of $450K, an average price of $540K, an average living area of 2K square feet and an average price of $264 per square foot.

The charts show average price by grade, bedrooms, waterfront status, bathrooms and condition. The map is colored by price tier, and the name and roll number are shown at the bottom. All visuals are on one page, so a manager can see the whole market without switching pages.

4. Key Observations from the Dashboard

- Market overview: about 22K houses were sold, with an average price of about $540K and a median of about $450K. The average is higher than the median, so a small number of expensive houses raise the average.
- Size and value: the average living area is about 2K square feet, and the average price is about $264 per square foot.
- Grade: average price rises steadily with property grade and climbs very sharply at the highest grades, so grade is one of the strongest drivers of price.
- Condition: houses in better condition tend to have a higher average price, although the difference between the lower conditions is small.
- Bedrooms: average price increases as the number of bedrooms increases up to about eight bedrooms, and then falls for larger counts.

- Waterfront: waterfront houses have a much higher average price than non-waterfront houses, so waterfront availability adds a large premium.
- Location: the map shows that most sales are concentrated on the western side of the county close to Seattle and the water, and the expensive tiers are found mainly in that area.

6. Conclusion

In this project I built the House Sales Analytics Dashboard in Power BI. I created five DAX measures, stored them in a separate measures table, added a Waterfront Status column, built a map of house locations colored by price tier, created price charts by grade, condition, bedrooms, bathrooms and waterfront status, and added Min SqFt and Max SqFt numeric range parameters as sliders.

The dashboard shows that price is strongly linked to grade, number of bedrooms, waterfront location and area in the county. It gives real-estate managers a quick and interactive view of the market and meets the objectives set for Week 10.